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CTA Station Distribution and Racial Composition in Chicago

Exploring the relationship between the distribution of CTA bus and railway stations amongst Chicago's community areas

Research Question and Background

How does access to CTA bus and rail transit vary across communities with different racial compositions in Chicago?

- Public transit is a critical public resource
- Distribution of transit infrastructure across cities is commonly uneven
- The history of segregation in Chicago could shape transit access
- Understanding these disparities can inform equitable planning and policy

Data Sources & Processing

1

Spatially joined CTA stops to community areas

2

Aggregated CTA stop counts by community area

3

Merged transit data with community level demographic data

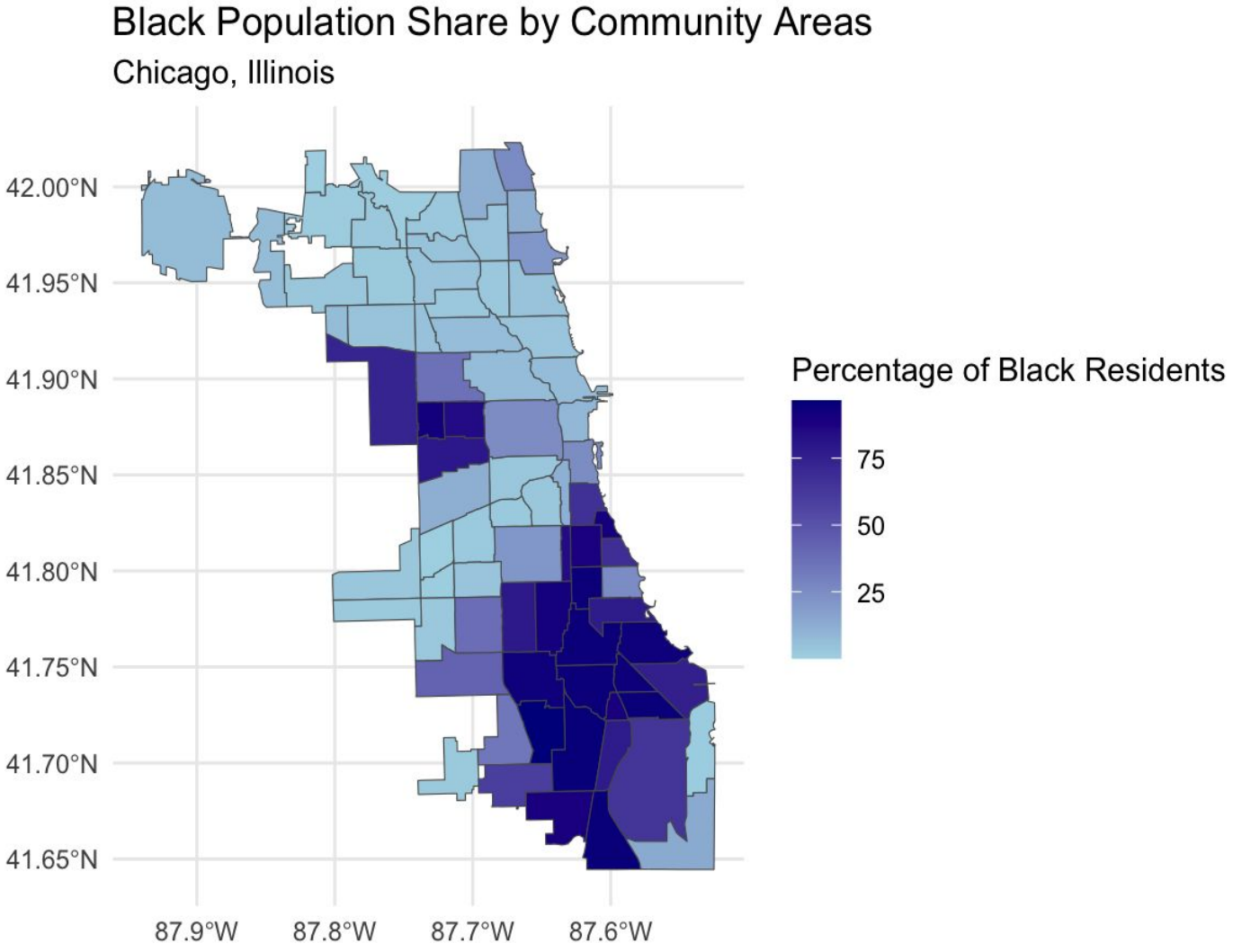
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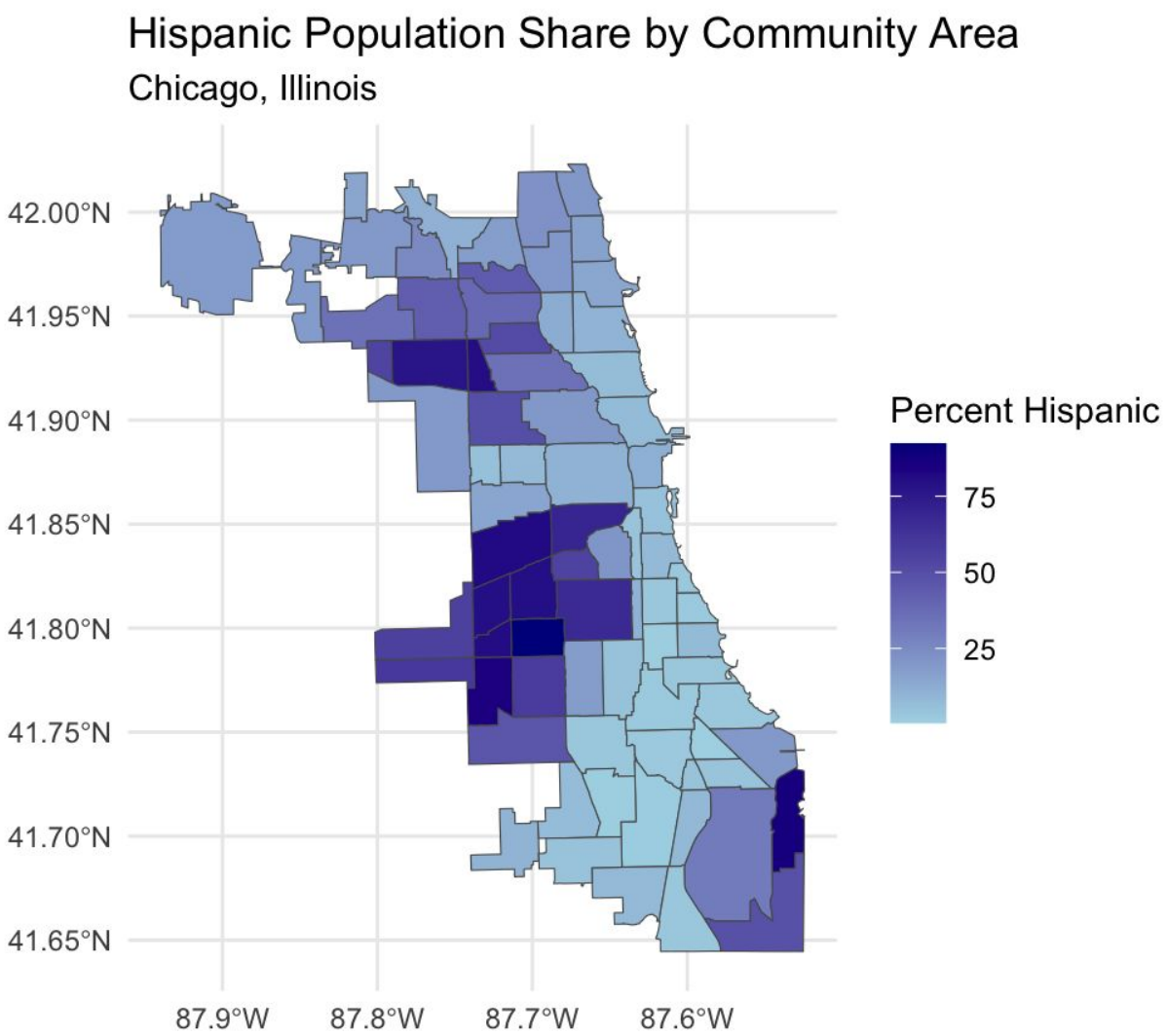
Created per capita measures:

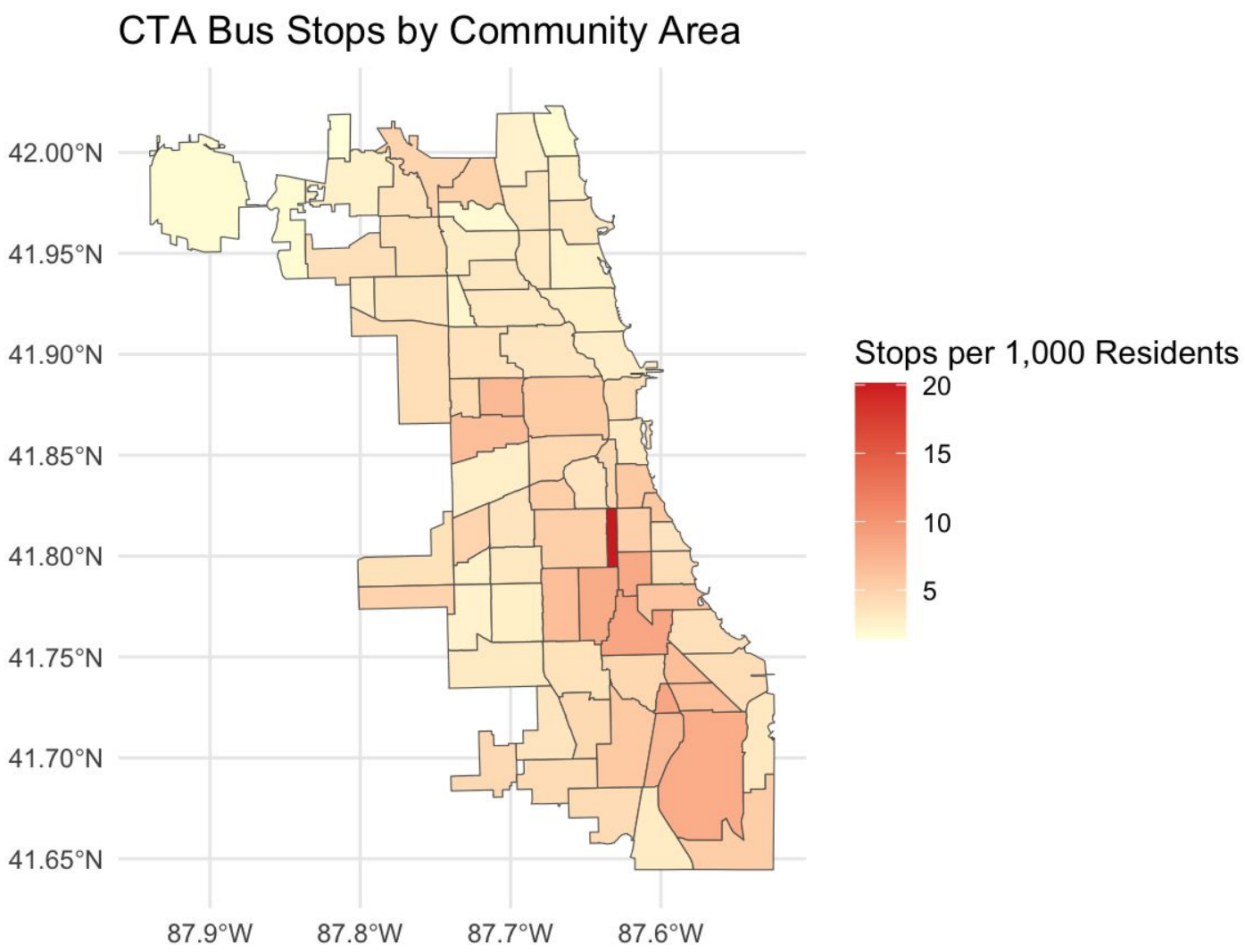
- Bus stops per 1,000 residents
- Train stops per 10,000 residents

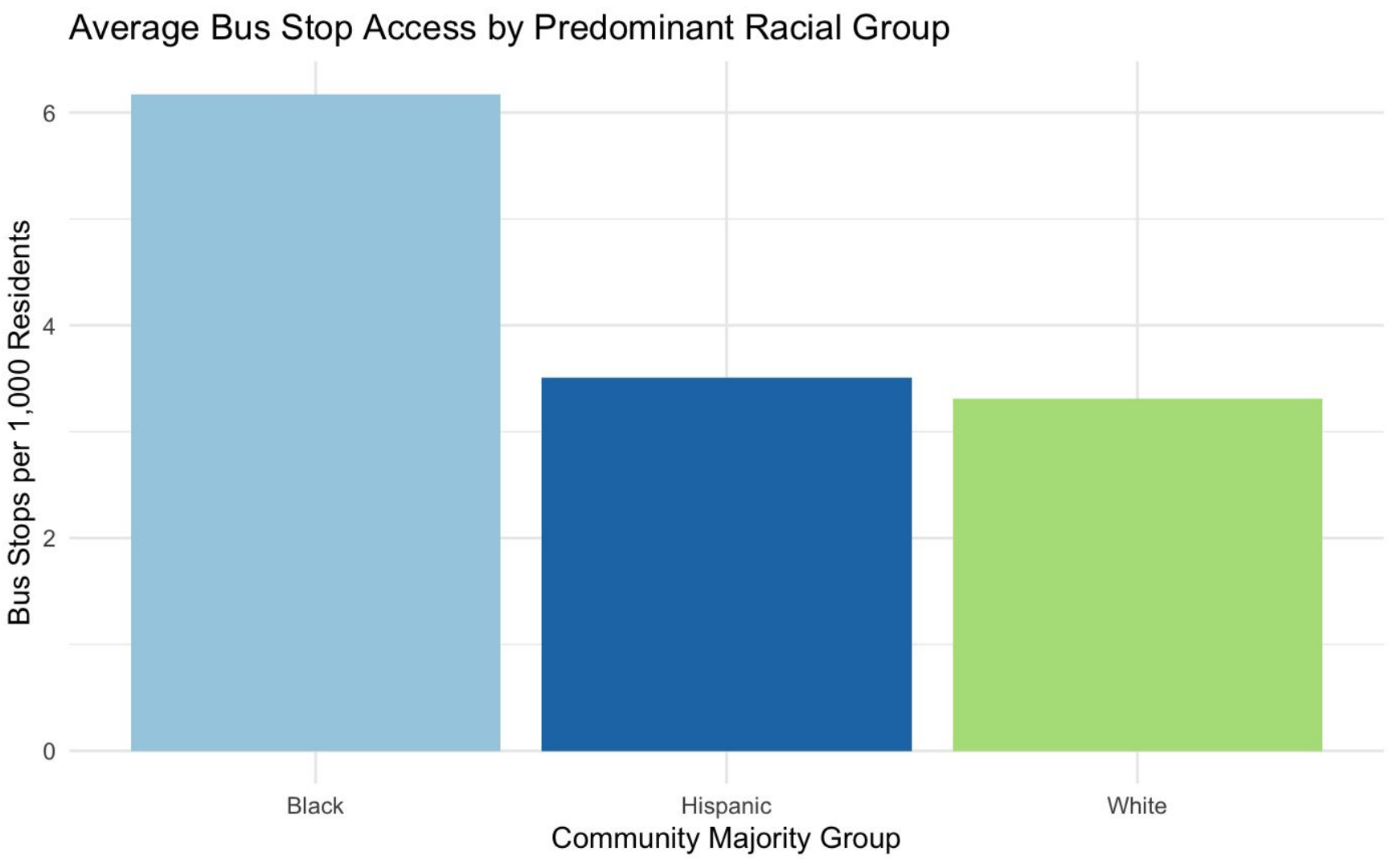
Sources

- Chicago ACS data
- CTA Bus Stop Locations
- CTA Rail Station Locations
- Chicago community area boundaries


Black Population Share by Community Areas
Chicago, Illinois
42.00°N
41.95°N
41.90°N
41.85°N
41.80°N
41.75°N
41.70°N
41.65°N
87.9°W
87.8°W
87.7°W
87.6°W
Percentage of Black Residents
75
50
25


Hispanic Population Share by Community Area
Chicago, Illinois
42.00°N
41.95°N
41.90°N
41.85°N
41.80°N
41.75°N
41.70°N
41.65°N
87.9°W
87.8°W
87.7°W
87.6°W
Percent Hispanic
75
50
25

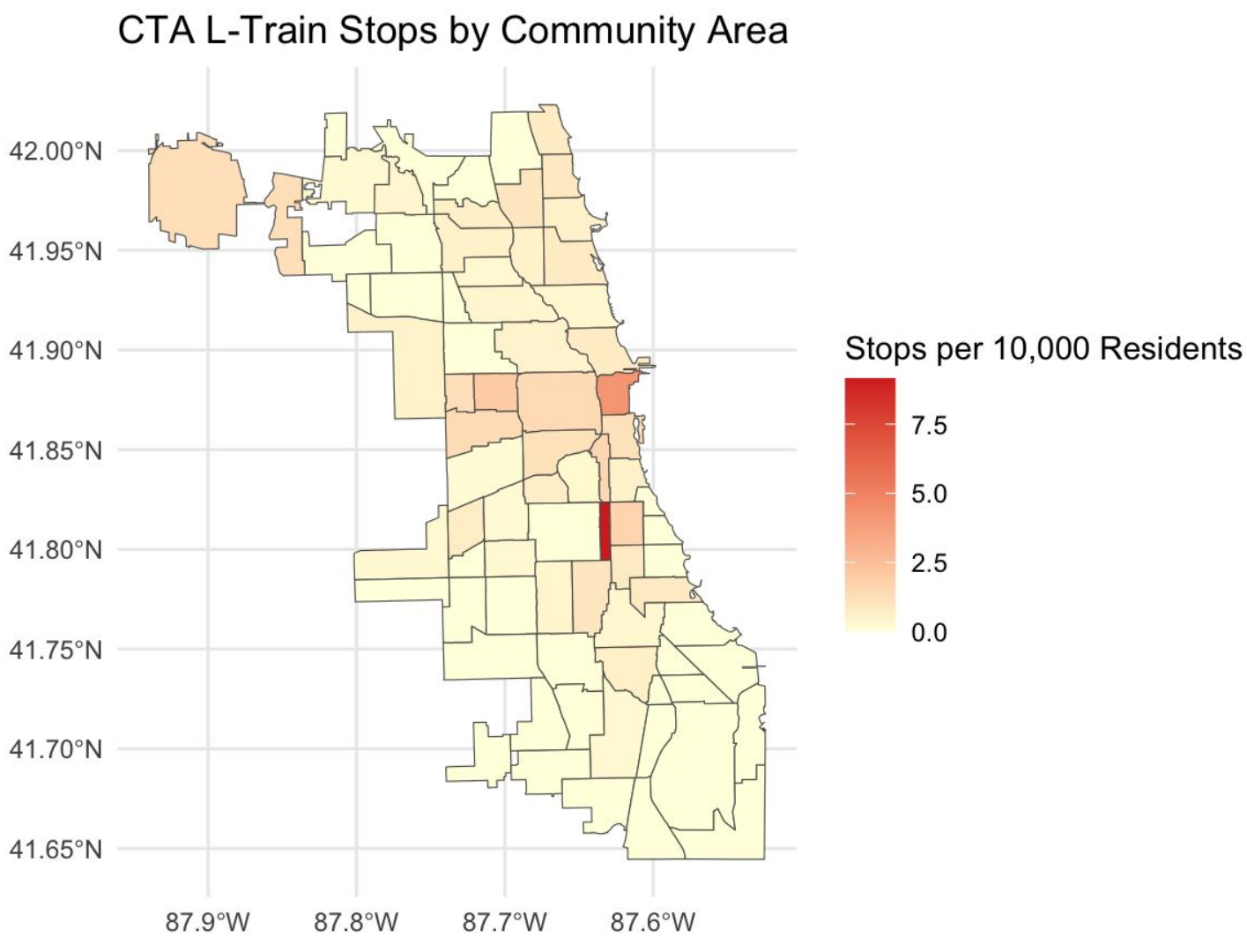

CTA Bus Stops by Community Area
42.00°N
41.95°N
41.90°N
41.85°N
41.80°N
41.75°N
41.70°N
41.65°N
87.9°W
87.8°W
87.7°W
87.6°W
Stops per 1,000 Residents
20
15
10
5

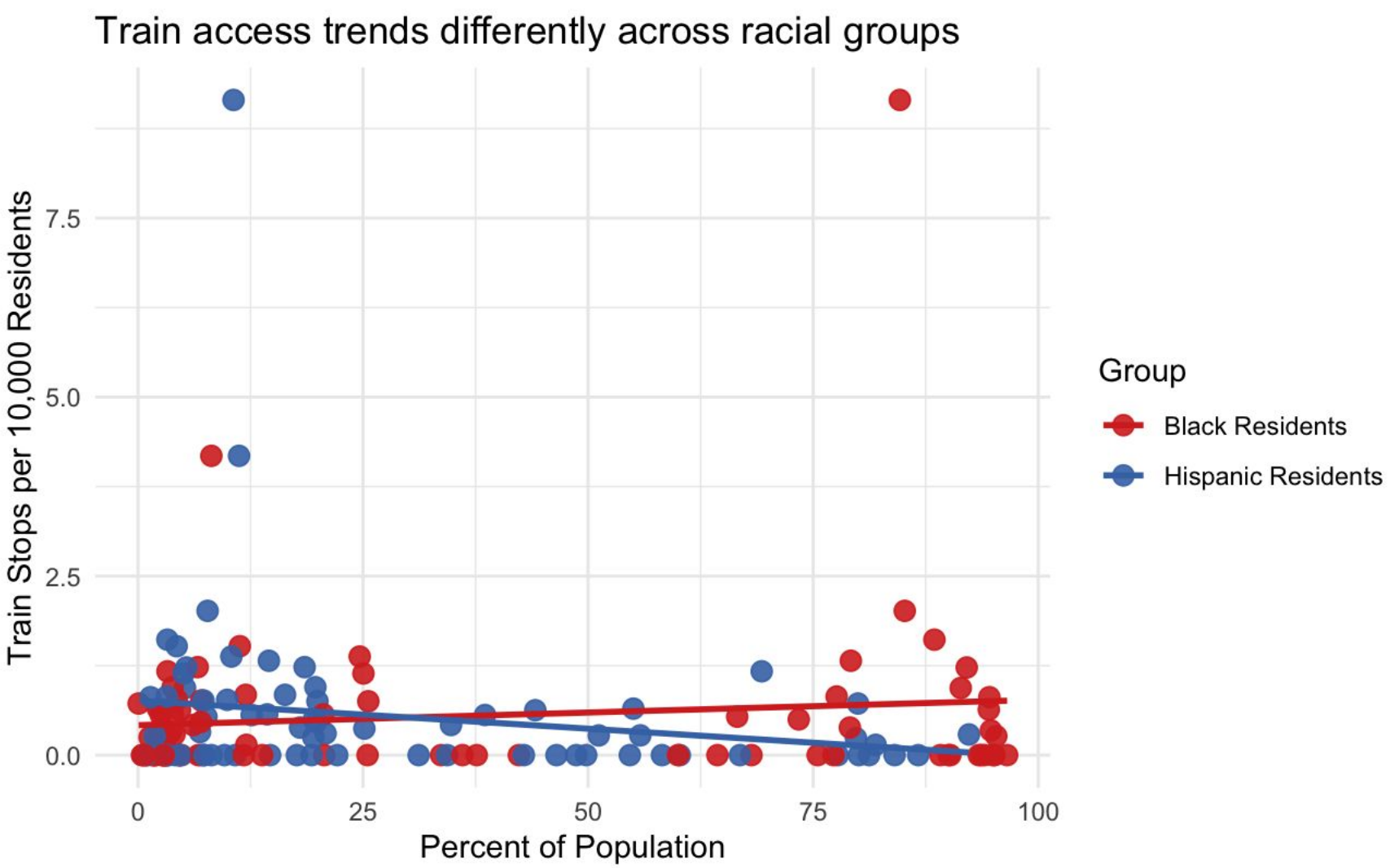

Average Bus Stop Access by Predominant Racial Group
Bus Stops per 1,000 Residents
6
4
2
0
Black
Hispanic
White
Community Majority Group

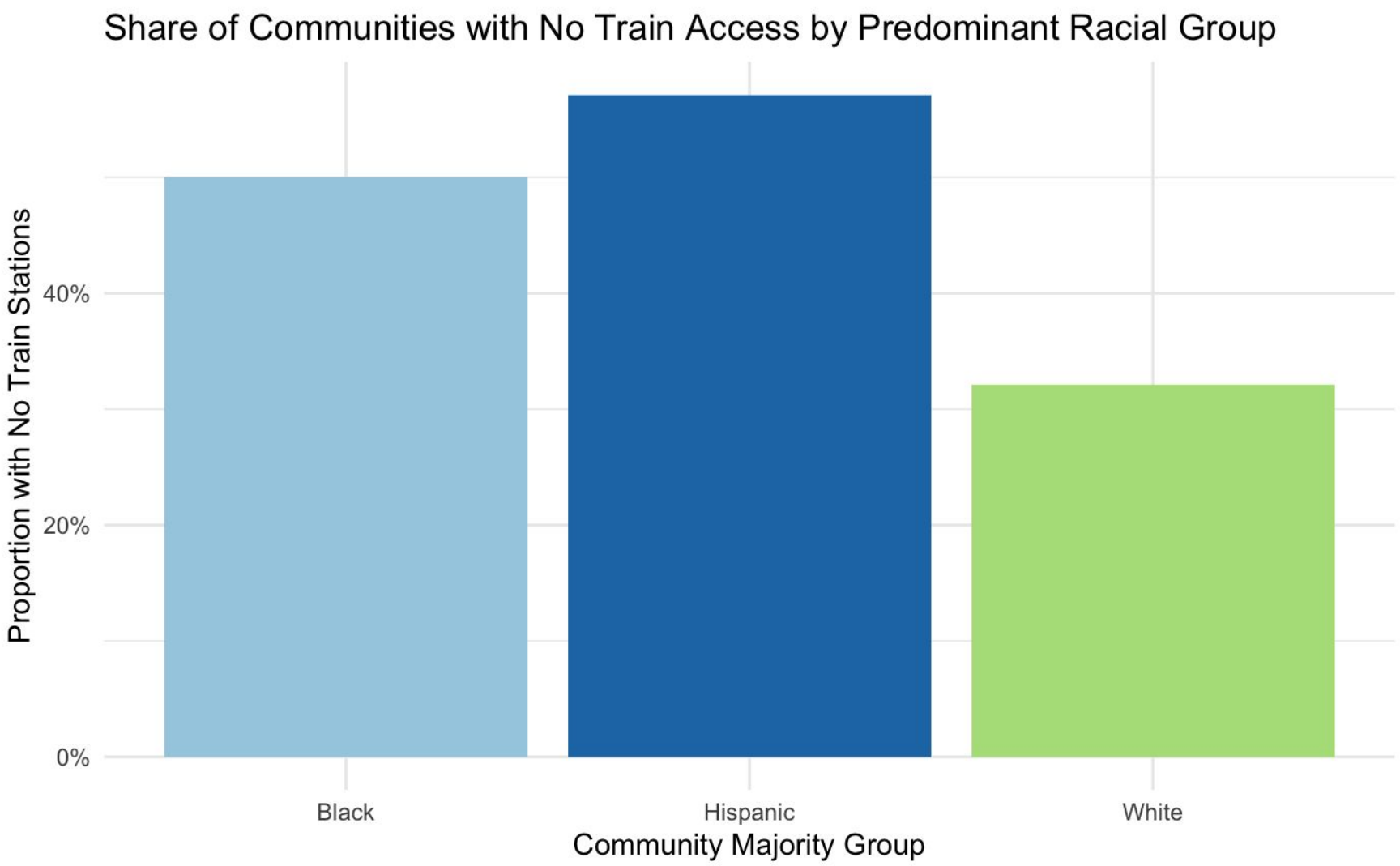
What this tells us about Bus access...

- There is a moderate positive association between bus stop distribution and predominantly Black/African American communities

- The distribution of bus stops is relatively consistent between predominantly white and Hispanic/Latino communities


CTA L-Train Stops by Community Area
42.00°N
41.95°N
41.90°N
41.85°N
41.80°N
41.75°N
41.70°N
41.65°N
87.9°W
87.8°W
87.7°W
87.6°W
Stops per 10,000 Residents
7.5
5.0
2.5
0.0


Train access trends differently across racial groups
Train Stops per 10,000 Residents
7.5
5.0
2.5
0.0
0
25
50
75
100
Percent of Population
Group
Black Residents
Hispanic Residents


Share of Communities with No Train Access by Predominant Racial Group
Proportion with No Train Stations
40%
20%
0%
Black
Hispanic
White
Community Majority Group

Overall Findings

- The spatial distribution of Chicago's population reflects patterns of racial segregation across community areas

- Bus stop access suggests a greater bus availability in predominantly Black communities, with substantial variation across neighborhoods

- CTA train stations are more concentrated in predominantly white areas, leaving many Black and Latino communities with little to no train access



Limitations

Age of Data

- Two datasets accessed through the Chicago data portal and used in this study were published in 2022

Exclusion of additional racial demographics

- This analysis only accounts for poplations within the Latino, Black/African American, and White communities

Frequency and reliability of accessible transit

- Red and Blue Lines operate on a 24hr basis while all other lines start and end service at specified times

Route Direction

- Stations may be present but certain routes may limit the access residents have to certain areas

Takeaways

Interpretation

- Accessibility differs by mode and location
- Uneven distributions of transit infrastructure suggest differences in connectivity and mobility opportunities

Policy Relevance

- Expansion of transit infrastructure could improve connectivity for underserved communities
- Equitable transit planning can shape access to jobs, services, and opportunity